Lesson Planning

Complete Lesson Plan Book

Class: Class 12 – FSc Pre-Engineering

Subject: Chemistry

Medium: English

Academic Year 2026-27

DM

This book is prepared as a teaching guide for teachers.

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Unit 1: Fundamental Principles of Organic Chemistry

Learning Objectives

• Define and explain hybridization in carbon in the context of Fundamental Principles of Organic Chemistry.

• Differentiate between isomerism and inductive effect with suitable examples.

• Solve numerical problems related to resonance accurately.

• Apply the ideas of Fundamental Principles of Organic Chemistry to everyday situations involving reaction mechanisms.

Key Concepts

• Meaning, definition and scope of hybridization in carbon.

• Important properties, types and examples of isomerism.

• Relationship between inductive effect and resonance.

• Practical applications of Fundamental Principles of Organic Chemistry, especially reaction mechanisms.

• Key terms and a summary of the main points of Fundamental Principles of Organic Chemistry.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Fundamental Principles of Organic Chemistry with a daily-life example.

2. Explain hybridization in carbon on the board with definitions and clear diagrams.

3. Discuss isomerism and inductive effect using simple, relatable examples.

4. Involve the students through short questions on resonance.

5. Solve a numerical on reaction mechanisms together with the class.

6. Summarize the lesson and highlight the key points of Fundamental Principles of Organic Chemistry.

Classroom Activities

• Group discussion: students compare isomerism with inductive effect.

• Board activity: solve a question on resonance in pairs.

• Quick quiz on the key terms of Fundamental Principles of Organic Chemistry.

Assessment Questions

1. Define hybridization in carbon.

2. Differentiate between isomerism and inductive effect.

3. Solve a numerical based on resonance.

4. Write a short note on reaction mechanisms.

Homework

• Read Fundamental Principles of Organic Chemistry from the book and solve the exercise questions.

• Prepare short notes on hybridization in carbon and isomerism.

Time Allocation

4–6 periods of 40 minutes each

Unit 2: Aliphatic Hydrocarbons

Learning Objectives

• Define and explain alkanes in the context of Aliphatic Hydrocarbons.

• Differentiate between alkenes and alkynes with suitable examples.

• Solve numerical problems related to Markovnikov's rule accurately.

• Apply the ideas of Aliphatic Hydrocarbons to everyday situations involving polymerization.

Key Concepts

• Meaning, definition and scope of alkanes.

• Important properties, types and examples of alkenes.

• Relationship between alkynes and Markovnikov's rule.

• Practical applications of Aliphatic Hydrocarbons, especially polymerization.

• Key terms and a summary of the main points of Aliphatic Hydrocarbons.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Aliphatic Hydrocarbons with a daily-life example.
2. Explain alkanes on the board with definitions and clear diagrams.
3. Discuss alkenes and alkynes using simple, relatable examples.
4. Involve the students through short questions on Markovnikov's rule.
5. Solve a numerical on polymerization together with the class.
6. Summarize the lesson and highlight the key points of Aliphatic Hydrocarbons.

Classroom Activities

• Group discussion: students compare alkenes with alkynes.

• Board activity: solve a question on Markovnikov's rule in pairs.

• Quick quiz on the key terms of Aliphatic Hydrocarbons.

Assessment Questions

1. Define alkanes.
2. Differentiate between alkenes and alkynes.
3. Solve a numerical based on Markovnikov's rule.
4. Write a short note on polymerization.

Homework

• Read Aliphatic Hydrocarbons from the book and solve the exercise questions.

• Prepare short notes on alkanes and alkenes.

Time Allocation

4–6 periods of 40 minutes each

Unit 3: Aromatic Hydrocarbons

Learning Objectives

• Define and explain benzene structure in the context of Aromatic Hydrocarbons.
• Differentiate between electrophilic substitution and nitration with suitable examples.
• Solve numerical problems related to sulphonation accurately.
• Apply the ideas of Aromatic Hydrocarbons to everyday situations involving uses of benzene.

Key Concepts

• Meaning, definition and scope of benzene structure.
• Important properties, types and examples of electrophilic substitution.
• Relationship between nitration and sulphonation.
• Practical applications of Aromatic Hydrocarbons, especially uses of benzene.
• Key terms and a summary of the main points of Aromatic Hydrocarbons.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Aromatic Hydrocarbons with a daily-life example.
2. Explain benzene structure on the board with definitions and clear diagrams.
3. Discuss electrophilic substitution and nitration using simple, relatable examples.
4. Involve the students through short questions on sulphonation.
5. Solve a numerical on uses of benzene together with the class.
6. Summarize the lesson and highlight the key points of Aromatic Hydrocarbons.

Classroom Activities

• Group discussion: students compare electrophilic substitution with nitration.
• Board activity: solve a question on sulphonation in pairs.
• Quick quiz on the key terms of Aromatic Hydrocarbons.

Assessment Questions

1. Define benzene structure.
2. Differentiate between electrophilic substitution and nitration.
3. Solve a numerical based on sulphonation.
4. Write a short note on uses of benzene.

Homework

• Read Aromatic Hydrocarbons from the book and solve the exercise questions.
• Prepare short notes on benzene structure and electrophilic substitution.

Time Allocation

4–6 periods of 40 minutes each

Unit 4: Alkyl Halides

Learning Objectives

• Define and explain nomenclature in the context of Alkyl Halides.

• Differentiate between SN1 reaction and SN2 reaction with suitable examples.

• Solve numerical problems related to elimination reactions accurately.

• Apply the ideas of Alkyl Halides to everyday situations involving Grignard reagent.

Key Concepts

• Meaning, definition and scope of nomenclature.

• Important properties, types and examples of SN1 reaction.

• Relationship between SN2 reaction and elimination reactions.

• Practical applications of Alkyl Halides, especially Grignard reagent.

• Key terms and a summary of the main points of Alkyl Halides.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Alkyl Halides with a daily-life example.
2. Explain nomenclature on the board with definitions and clear diagrams.
3. Discuss SN1 reaction and SN2 reaction using simple, relatable examples.
4. Involve the students through short questions on elimination reactions.
5. Solve a numerical on Grignard reagent together with the class.
6. Summarize the lesson and highlight the key points of Alkyl Halides.

Classroom Activities

• Group discussion: students compare SN1 reaction with SN2 reaction.

• Board activity: solve a question on elimination reactions in pairs.

• Quick quiz on the key terms of Alkyl Halides.

Assessment Questions

1. Define nomenclature.
2. Differentiate between SN1 reaction and SN2 reaction.
3. Solve a numerical based on elimination reactions.
4. Write a short note on Grignard reagent.

Homework

• Read Alkyl Halides from the book and solve the exercise questions.

• Prepare short notes on nomenclature and SN1 reaction.

Time Allocation

4–6 periods of 40 minutes each

Unit 5: Alcohols and Phenols

Learning Objectives

• Define and explain classification in the context of Alcohols and Phenols.
• Differentiate between preparation methods and oxidation with suitable examples.
• Solve numerical problems related to acidity of phenol accurately.
• Apply the ideas of Alcohols and Phenols to everyday situations involving uses.

Key Concepts

• Meaning, definition and scope of classification.
• Important properties, types and examples of preparation methods.
• Relationship between oxidation and acidity of phenol.
• Practical applications of Alcohols and Phenols, especially uses.
• Key terms and a summary of the main points of Alcohols and Phenols.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Alcohols and Phenols with a daily-life example.
2. Explain classification on the board with definitions and clear diagrams.
3. Discuss preparation methods and oxidation using simple, relatable examples.
4. Involve the students through short questions on acidity of phenol.
5. Solve a numerical on uses together with the class.
6. Summarize the lesson and highlight the key points of Alcohols and Phenols.

Classroom Activities

• Group discussion: students compare preparation methods with oxidation.
• Board activity: solve a question on acidity of phenol in pairs.
• Quick quiz on the key terms of Alcohols and Phenols.

Assessment Questions

1. Define classification.
2. Differentiate between preparation methods and oxidation.
3. Solve a numerical based on acidity of phenol.
4. Write a short note on uses.

Homework

• Read Alcohols and Phenols from the book and solve the exercise questions.
• Prepare short notes on classification and preparation methods.

Time Allocation

4–6 periods of 40 minutes each

Unit 6: Aldehydes and Ketones

Learning Objectives

• Define and explain nomenclature in the context of Aldehydes and Ketones.
• Differentiate between nucleophilic addition and oxidation tests with suitable examples.
• Solve numerical problems related to aldol condensation accurately.
• Apply the ideas of Aldehydes and Ketones to everyday situations involving uses.

Key Concepts

• Meaning, definition and scope of nomenclature.
• Important properties, types and examples of nucleophilic addition.
• Relationship between oxidation tests and aldol condensation.
• Practical applications of Aldehydes and Ketones, especially uses.
• Key terms and a summary of the main points of Aldehydes and Ketones.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Aldehydes and Ketones with a daily-life example.
2. Explain nomenclature on the board with definitions and clear diagrams.
3. Discuss nucleophilic addition and oxidation tests using simple, relatable examples.
4. Involve the students through short questions on aldol condensation.
5. Solve a numerical on uses together with the class.
6. Summarize the lesson and highlight the key points of Aldehydes and Ketones.

Classroom Activities

• Group discussion: students compare nucleophilic addition with oxidation tests.
• Board activity: solve a question on aldol condensation in pairs.
• Quick quiz on the key terms of Aldehydes and Ketones.

Assessment Questions

1. Define nomenclature.
2. Differentiate between nucleophilic addition and oxidation tests.
3. Solve a numerical based on aldol condensation.
4. Write a short note on uses.

Homework

• Read Aldehydes and Ketones from the book and solve the exercise questions.
• Prepare short notes on nomenclature and nucleophilic addition.

Time Allocation

4–6 periods of 40 minutes each

Unit 7: Carboxylic Acids

Learning Objectives

• Define and explain nomenclature in the context of Carboxylic Acids.
• Differentiate between acidity and esterification with suitable examples.
• Solve numerical problems related to derivatives accurately.
• Apply the ideas of Carboxylic Acids to everyday situations involving uses.

Key Concepts

• Meaning, definition and scope of nomenclature.
• Important properties, types and examples of acidity.
• Relationship between esterification and derivatives.
• Practical applications of Carboxylic Acids, especially uses.
• Key terms and a summary of the main points of Carboxylic Acids.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Carboxylic Acids with a daily-life example.
2. Explain nomenclature on the board with definitions and clear diagrams.
3. Discuss acidity and esterification using simple, relatable examples.
4. Involve the students through short questions on derivatives.
5. Solve a numerical on uses together with the class.
6. Summarize the lesson and highlight the key points of Carboxylic Acids.

Classroom Activities

• Group discussion: students compare acidity with esterification.
• Board activity: solve a question on derivatives in pairs.
• Quick quiz on the key terms of Carboxylic Acids.

Assessment Questions

1. Define nomenclature.
2. Differentiate between acidity and esterification.
3. Solve a numerical based on derivatives.
4. Write a short note on uses.

Homework

• Read Carboxylic Acids from the book and solve the exercise questions.
• Prepare short notes on nomenclature and acidity.

Time Allocation

4–6 periods of 40 minutes each

Unit 8: Macromolecules

Learning Objectives

• Define and explain polymers in the context of Macromolecules.

• Differentiate between addition polymerization and condensation polymerization with suitable examples.

• Solve numerical problems related to proteins accurately.

• Apply the ideas of Macromolecules to everyday situations involving nylon and polyester.

Key Concepts

• Meaning, definition and scope of polymers.

• Important properties, types and examples of addition polymerization.

• Relationship between condensation polymerization and proteins.

• Practical applications of Macromolecules, especially nylon and polyester.

• Key terms and a summary of the main points of Macromolecules.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Macromolecules with a daily-life example.
2. Explain polymers on the board with definitions and clear diagrams.
3. Discuss addition polymerization and condensation polymerization using simple, relatable examples.
4. Involve the students through short questions on proteins.
5. Solve a numerical on nylon and polyester together with the class.
6. Summarize the lesson and highlight the key points of Macromolecules.

Classroom Activities

• Group discussion: students compare addition polymerization with condensation polymerization.

• Board activity: solve a question on proteins in pairs.

• Quick quiz on the key terms of Macromolecules.

Assessment Questions

1. Define polymers.
2. Differentiate between addition polymerization and condensation polymerization.
3. Solve a numerical based on proteins.
4. Write a short note on nylon and polyester.

Homework

• Read Macromolecules from the book and solve the exercise questions.

• Prepare short notes on polymers and addition polymerization.

Time Allocation

4–6 periods of 40 minutes each

Unit 9: Common Chemical Industries

Learning Objectives

• Define and explain urea manufacturing in the context of Common Chemical Industries.

• Differentiate between cement industry and soap and detergents with suitable examples.

• Solve numerical problems related to paper industry accurately.

• Apply the ideas of Common Chemical Industries to everyday situations involving petroleum refining.

Key Concepts

• Meaning, definition and scope of urea manufacturing.

• Important properties, types and examples of cement industry.

• Relationship between soap and detergents and paper industry.

• Practical applications of Common Chemical Industries, especially petroleum refining.

• Key terms and a summary of the main points of Common Chemical Industries.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Common Chemical Industries with a daily-life example.
2. Explain urea manufacturing on the board with definitions and clear diagrams.
3. Discuss cement industry and soap and detergents using simple, relatable examples.
4. Involve the students through short questions on paper industry.
5. Solve a numerical on petroleum refining together with the class.
6. Summarize the lesson and highlight the key points of Common Chemical Industries.

Classroom Activities

• Group discussion: students compare cement industry with soap and detergents.

• Board activity: solve a question on paper industry in pairs.

• Quick quiz on the key terms of Common Chemical Industries.

Assessment Questions

1. Define urea manufacturing.
2. Differentiate between cement industry and soap and detergents.
3. Solve a numerical based on paper industry.
4. Write a short note on petroleum refining.

Homework

• Read Common Chemical Industries from the book and solve the exercise questions.

• Prepare short notes on urea manufacturing and cement industry.

Time Allocation

4–6 periods of 40 minutes each

Unit 10: Environmental Chemistry

Learning Objectives

• Define and explain air pollution in the context of Environmental Chemistry.
• Differentiate between water pollution and ozone depletion with suitable examples.
• Solve numerical problems related to acid rain accurately.
• Apply the ideas of Environmental Chemistry to everyday situations involving green chemistry.

Key Concepts

• Meaning, definition and scope of air pollution.
• Important properties, types and examples of water pollution.
• Relationship between ozone depletion and acid rain.
• Practical applications of Environmental Chemistry, especially green chemistry.
• Key terms and a summary of the main points of Environmental Chemistry.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Environmental Chemistry with a daily-life example.
2. Explain air pollution on the board with definitions and clear diagrams.
3. Discuss water pollution and ozone depletion using simple, relatable examples.
4. Involve the students through short questions on acid rain.
5. Solve a numerical on green chemistry together with the class.
6. Summarize the lesson and highlight the key points of Environmental Chemistry.

Classroom Activities

• Group discussion: students compare water pollution with ozone depletion.
• Board activity: solve a question on acid rain in pairs.
• Quick quiz on the key terms of Environmental Chemistry.

Assessment Questions

1. Define air pollution.
2. Differentiate between water pollution and ozone depletion.
3. Solve a numerical based on acid rain.
4. Write a short note on green chemistry.

Homework

• Read Environmental Chemistry from the book and solve the exercise questions.
• Prepare short notes on air pollution and water pollution.

Time Allocation

4–6 periods of 40 minutes each